

M. Tech Semester-III (Mid Semester) Examination, 2025

Subject: Computer Science and Engineering

Paper Code & Name: CSOE-11XX: Open Elective (Operation Research)

Full Marks: 20

Date: 10.12.2025

Time and Duration: 12:00-1:00 PM; 1 hr.

Answer any two from Q.1 to Q3 and Q4 is Mandatory

1. Solve the Transportation problem using VAM method and also perform the optimality test of the given problem:

7.5

					Source
	2	3	11	7	6
	1	0	6	1	1
	5	8	15	9	10
Destination	7	5	3	2	

2. Solve the Transportation problem using VAM method and also perform the optimality test of the given problem:

7.5

						Source
	7	6	4	5	9	40
	8	5	6	7	8	30
	6	8	9	6	5	20
	5	7	7	8	6	10
Destination	30	30	15	20	5	

3. Use the dominance rule to reduce the payoff matrix and solve the game:

7.5

	B1	B2	B3	B4
A1	8	15	-4	-2
A2	19	15	17	16
A3	0	20	15	5

4. Solve the following Assignment Problem:

5

	O1	O2	O3	O4	O5
J1	10	12	15	12	8
J2	7	16	14	14	11
J3	13	14	7	9	9
J4	12	10	11	13	10
J5	8	13	15	11	15